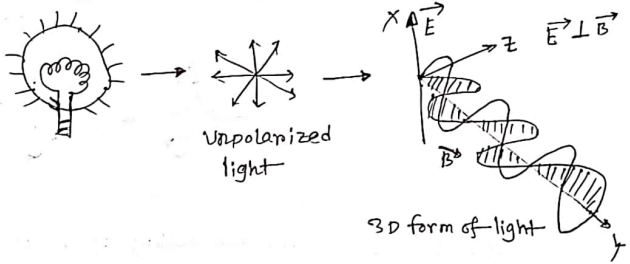
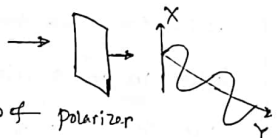
Chap-05

Introduction and Basic principles of ORD

Optical Rotatory Dispersion

Light ⇒ Electromagnetic Radiation





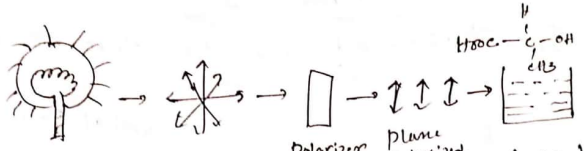
Generally 2 types of polarizer crystal ⇒

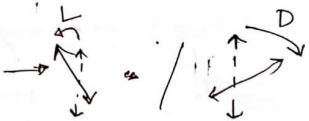
- Glan-Thomson prism (Tourmaline crystal) — Most uses due to its betterment of polarizer
- Nicu prism (Calcite crystal)

plane polarized light

Unpolarized Light → ↕ ↕ ↕ ↕ ↕

□ → ↕ ↕ ↕ ↕ ↕



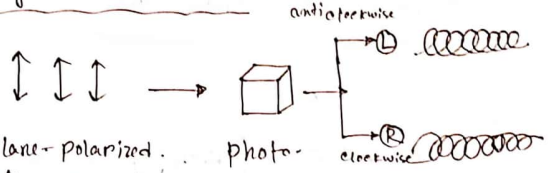


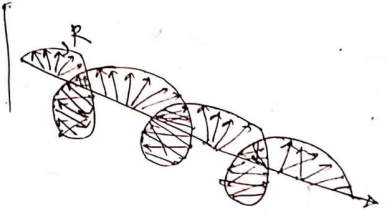
ORD : ORD measures the ability of an optically active compound to rotate plane polarized light in an anticlockwise or clockwise direction with change in the wavelength of light.

The instrument which is used to measure ORD is known as spectropolarimeter.

Q:- What is circularly polarized light and plane polarized light? with figure.

Circularly polarized light :



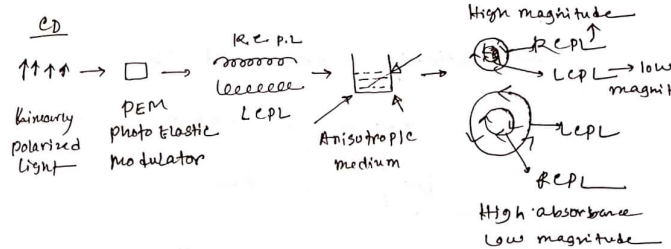


When plane polarized light incident on the Photo elastic modulator (PEM), then PEM chang the plane polarized light into circularly polarized light which is known as circularly polarized light.

Circularly polarized light is a combination of left and right circularly polarized light.

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CD



PEM incident different form of (Right / Left) circularly polarized light on a anisotropic medium alternatively.

PEM does not incident (left / right) C.P.L both at a single time.

When a L.C.P.L / R.C.P.L incident on an anisotropic medium, it absorbs the light which is not equal, the anisotropic medium can absorb high amount of LCPL and low amount of RCPL & vice versa.

Circular dichroism spectroscopy measures the difference in absorbance of RCPL & LCPL which arises due to structural assymetry of anisotically active substance.

Mathmatically Expression of CD

According to Beer Lamber law –

$$\log \frac{I}{I_0} = A = \varepsilon c l$$

Where, A = Absorbance

c = concentration (molarr).

l = path length of the cell

ε = Molar extinction coefficient

For CD,

$A_L \neq A_R$

$A_L = \varepsilon_L c l$, $A_R = \varepsilon_R c l$

$\therefore \Delta A = A_L - A_R$

$= \varepsilon_L c l - \varepsilon_R c l$

$= \Delta\varepsilon\, c l$

$\therefore \Delta\varepsilon = \frac{\Delta A}{cl} = \frac{\Delta A}{c}$ for unit length

Where, $\Delta\varepsilon$ is the molar CD

This intrinsic property is about usually means by the CD of the substance.

Since $\Delta\varepsilon$ is a function of wavelength, A molar CD ($\Delta\varepsilon$) value much specify the wavelength at which it is valid.

Application of CD

(i) used to determine the 2° structure of protein (β/α)

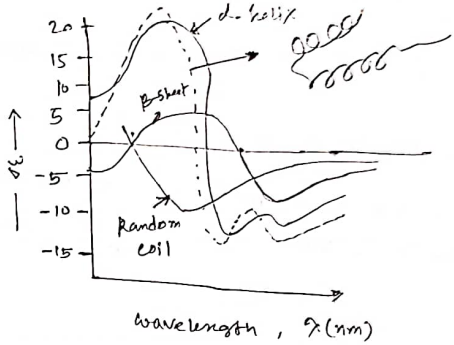
(ii) To study folding and unfolding of protein.

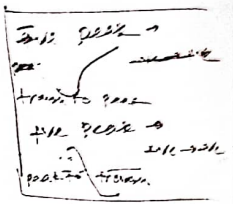
(iii) used to measure the structural changes made by heat, pH, solvent in protein

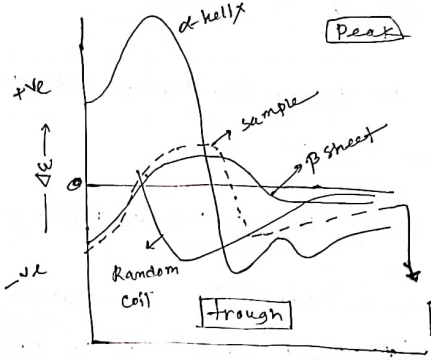
(iv) To study the structural changes of hormones protein, polysaccharides.

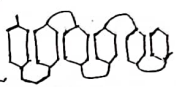
(v) Ligand — Binding changes in protein's

(*)









β-pleated Sheet

Q:- How do you understand by absorbing curve. to whether it is α-helix or β pleated sheet in protein?

Suppose that an anisotropic medium absorbs high amount of L.C.P.L and loss amount of R.C.P.L.

Due to high absorption of L.C.P.L the magnitude after absorption of L.C.P.L will be lower and the magnitude of R.C.P.L will be higher and vice-versa.

There are several molecules in an anisotropic medium which are ~~meaning~~ all the in different direction. Due to ~~rotent~~ movement the point of incident of C.P.L is diffent and absorption changed.

α-~~ket~~ Axial haloketone Rule :



This rule is applied for keto-steroid that is axially substituted with a halogen at C-α. Axial position is preferred due to dipole-dipole repulsion in equatorial. An axial halogen (Cl/Br/I) is to the α-position may alter the sign of the cotton effect.



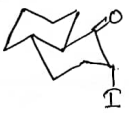
+ve cotton effect



+ve cotton effect



+ve CE.



-ve CE.

Cotton effect :

The cotton effect is the characteristic change in optical rotatory dispersion or circular dichroism in the vicinity of an absorption band of a substance. These curves will show the high peaks and troughs which depends on the absorbing groups. These curves will obtain for compounds which are having assymetric carbon and chromophoric groups.

CB. (Circular Bireferengence) :

The difference between n_L (Refractive index of left circularly polarized light) and n_R (Refractive index of right circularly polarized light) is called CB.

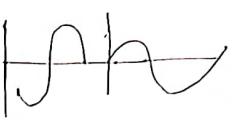
Cotton effect

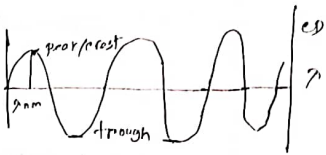
(+) CE

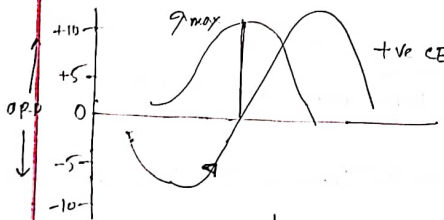
(−) CE

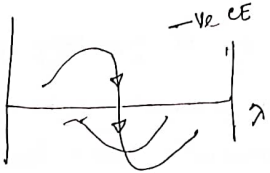
Single CE

Multiple CE











Single CE :

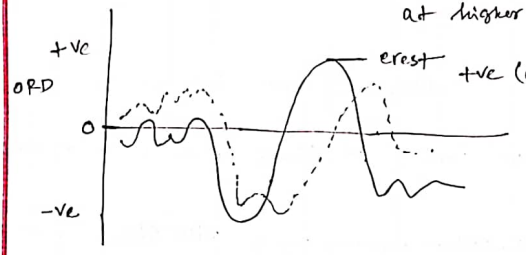
Single CE curves will show both maximum and minimum curves at λ_{max} or maximum absorption.

Multiple cotton effect curves will show several troughs and crests due to the presence of different groups in a compound.

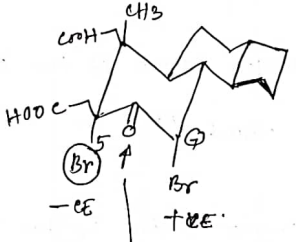
"At higher wavelength region if the crest comes first then it is called (+ve) CE."

"At higher wavelength region if the trough comes first, it is (+) CE."

Q: Circular Birefringence ?
CE ?, ORD & CD ?

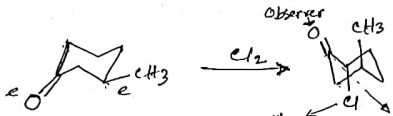


Application of ~~CH~~ α-Haloketone Rule :



→ With respect to observer, 5 no. Br
is left side → -CE
Right side → +CE

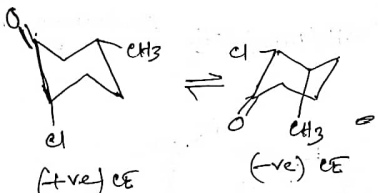
Chlorination of R(+)-3-methyl cyclohexane :



(As Cl group is on left side w.r. to observer)



Here, Cl / Br / I can be here but F is not due to its lower polarisability

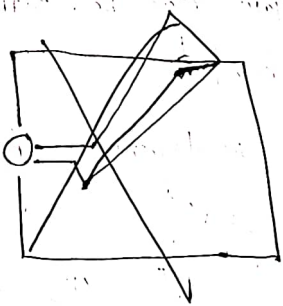


Octant Rule :

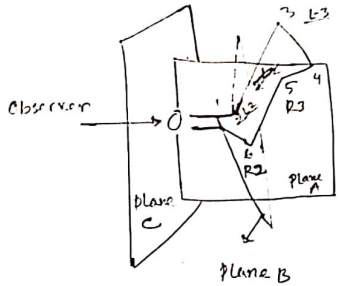
This rule is applied for substituted cyclo-hexanones for assigning configuration based on this rule. This rule is to devide the space around the carbonyl group into ~~to~~ eight sectors (octant), viewing will be through the $yz=0$ and this octant rule has two sectors

(I) Front sectors 

(II) Rear sector  Important



4th → in the order of planes
L → left referred to the observer
R → Right



Plane A → c=0, -c-4

Plane B → c-1, c-2, c-4

Plane c → perpendicular to c=0

Front sector

1st sector

Plane



Towards the observer → A → (+)

Right to the observer → B → (+) → (+)

Above the plane → C → (+)

2nd sector

A. Front (+)
B. Left (-)
C. Above (+)
→ (-)

Similarly,
3rd sector → (+)
4th sector → (-)

3rd sector :

Front {A} → (+)
Left {B} → (−)
Below {C} → (−)
} → (+)

4th sector :

Front · A → (+)
Right B → (+)
Below C → (−)
} → (−)

Q: Sign determining of 3D box –

(2) Contribution of the substituents in different sector towards Cotton effect (CE) :–

(1) Contribution from hydrogens are usually ignored.
↳ no contribution

(2) Substituents at 4th position will have no effect on the CE. Since both equatorial and axial are on the nodal plane.

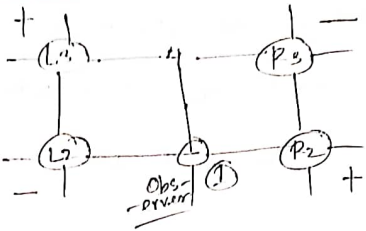
(3) Equatorial groups at 2nd and 6th position will make only small contribution to CE because the proximity to the XY plane.

(4) 2nd and 6th axial groups contribute more to CE

(5) Axial and equatorial substituents of 5th and 3rd position contributes equally

(6) Substituents at 5th and 2nd position make (+) CE

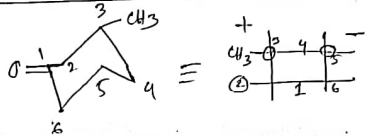
∴ " " 3rd and 6th " " (-) CE

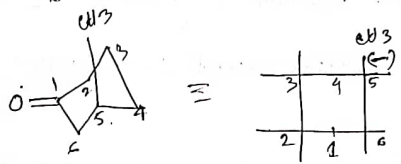


Rear

Q.11 (HA) Exam

Predetermination of conformation of known configuration by octant Rule :—

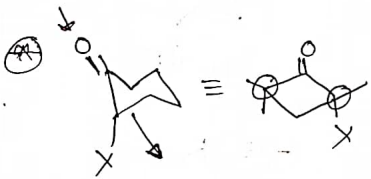




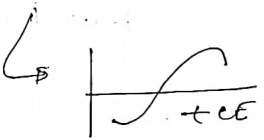
The compound R (+) methylcyclohexanone exhibit (+) CE.

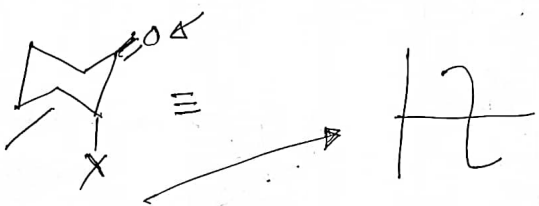
From the application of the octant rule

indicated that preffered conformation is equatorial. If the methyl is in equatorial position then (+) CE is recorded axial conformation is not ~~posit~~ possible.



(Right) (+ CE) → S-configuration





left (−CE) → R- configuration

(*)



(+). trans-2-decalone

Br_2



or



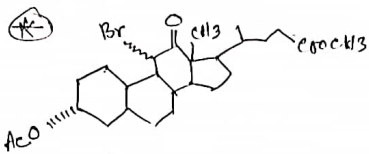
(+ cl) (S)

(Formed)

(- cl) (R)

Not formed experimentally

(*)



Br_2



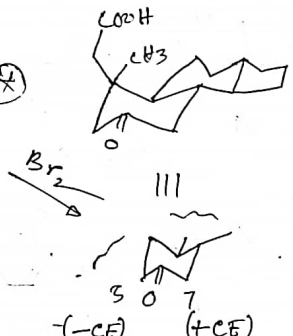


→ In this enantiomer Br is on the right side, it make (+ cl) So, we get (S-) config.

D.T.O

Fig 1.4

In this enantiomer, Br is on the left side, it makes −CE, so this enantiomer is possible as we get R− configuration.



Br → 5th position → −CE

Br → 7th position → +CE